Lecture 26 (*April 23*)

Reading: Section 6.4–6.5
Viewing: Lecture 25

Let us now consider eigenvalues and eigenvectors of symmetric matrices.

Fact 21. *A $n \times n$ symmetric matrix S has n real eigenvalues and n orthonormal eigenvectors.*

You can find a proof of the fact that the eigenvalues are real in the textbook, the bottom of page 339. For example, in the 2×2 case, a symmetric matrix:

$$S = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \tag{230}$$

has characteristic polynomial:

$$p(\lambda) = \lambda^2 - \lambda(a+c) + ac - b^2$$

whose roots are:

$$\frac{a + c \pm \sqrt{(a+c)^2 - 4ac + 4b^2}}{2} = \frac{a + c \pm \sqrt{(a-c)^2 + 4b^2}}{2} \tag{231}$$

It is clear that these are real, because $(a-c)^2 + 4b^2 \geq 0$. As for the fact that the eigenvectors of a symmetric matrix S are orthogonal, suppose $\lambda \neq \lambda'$ are distinct eigenvalues with corresponding eigenvectors $\boldsymbol{v}, \boldsymbol{v}'$. Then:

$$\lambda \cdot \boldsymbol{v}^T \boldsymbol{v}' = (S\boldsymbol{v})^T \boldsymbol{v}' = \boldsymbol{v}^T S^T \boldsymbol{v}' = \boldsymbol{v}^T S \boldsymbol{v}' = \lambda' \cdot \boldsymbol{v}^T \boldsymbol{v}'$$

which implies $\boldsymbol{v}^T \boldsymbol{v}' = 0$. Once you know that the eigenvectors are orthogonal, they can be made orthonormal by rescaling them appropriately (since any multiple of an eigenvector is an eigenvector). As a consequence of the fact above, we may put the orthonormal eigenvectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ in a matrix:

$$Q = \left[\ \boldsymbol{q}_1 \mid \ldots \mid \boldsymbol{q}_n\ \right]$$

which is, by definition, an orthogonal matrix. Then (191) implies that we may write:

$$S = Q \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{bmatrix} Q^{-1} = Q \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{bmatrix} Q^T \tag{232}$$

The last equality follows from the fact that $Q^{-1} = Q^T$, a general feature of orthogonal matrices.

Geometrically, equality (232) can be interpreted as the fact that the linear transformation corresponding to a symmetric matrix can always be factored as:

$$\Big(\text{rotation}\Big)\Big(\text{scaling}\Big)\Big(\text{inverse rotation}\Big) \tag{233}$$

Indeed, diagonal matrices correspond to scaling in the direction of the coordinate axes, while orthogonal matrices correspond to rotations in n–dimensional space (as an exercise, prove that any

2×2 orthogonal matrix must be literally equal to one of the rotation matrices from Lecture 16). Explicitly, formula (232) implies that we can always express a symmetric matrix as:

$$\boxed{S = \lambda_1 \boldsymbol{q}_1 \boldsymbol{q}_1^T + \cdots + \lambda_n \boldsymbol{q}_n \boldsymbol{q}_n^T} \tag{234}$$

(you can prove this by applying both sides of identity (234) to an arbitrary eigenvector $\boldsymbol{q}_i$, and then extending by linear combinations to arbitrary vectors).

Remark. *For an arbitrary (not necessarily symmetric) matrix A, the eigenvalues can be complex numbers. But whenever this happens, the following principle holds:*

$$\textbf{\textit{The complex eigenvalues of any real matrix come in conjugate pairs}} \tag{235}$$

For example, in a 2×2 example, the characteristic polynomial:

$$p(\lambda) = \lambda^2 - \lambda \cdot \mathrm{Tr} + \det$$

has roots:

$$\frac{\mathrm{Tr} \pm \sqrt{\mathrm{Tr}^2 - 4 \cdot \det}}{2}$$

If $\mathrm{Tr}^2 < 4 \cdot \det$, the roots are complex numbers, and they are manifestly conjugates of each other. Moreover, the eigenvectors corresponding to conjugate eigenvalues are also conjugate:

$$A\boldsymbol{v} = \lambda \boldsymbol{v} \qquad \Rightarrow \qquad A\overline{\boldsymbol{v}} = \bar{\lambda}\overline{\boldsymbol{v}}$$

While in general, the eigenvalues and pivots of a square matrix are not directly related to each other (other than the fact that the eigenvalues and the pivots both have the same product, i.e. the determinant), for a symmetric matrix we have:

$$\text{there are as many positive eigenvalues as positive pivots} \tag{236}$$

The analogous statement is true if "positive" were replaced by "negative".

Definition 21. *A symmetric matrix with all eigenvalues/pivots > 0 is called* ***positive definite****. A symmetric matrix with all eigenvalues/pivots ≥ 0 is called* ***positive semidefinite****.*

In general, if S is a positive definite (respectively semidefinite) matrix, then the "energy" [4]:

$$\boxed{\boldsymbol{v}^T S \boldsymbol{v} > 0} \qquad (\text{respectively } \boldsymbol{v}^T S \boldsymbol{v} \geq 0) \tag{237}$$

for any vector $\boldsymbol{v} \neq 0$. Indeed, just decompose $\boldsymbol{v}$ as a linear combination of the eigenvectors of S:

$$\boldsymbol{v} = c_1 \boldsymbol{q}_1 + \cdots + c_n \boldsymbol{q}_n$$

and then (237) becomes (because the $\boldsymbol{q}_i$'s are orthonormal):

$$\boldsymbol{v}^T S \boldsymbol{v} = \sum_{1 \leq i,j \leq n} c_i c_j \boldsymbol{q}_i^T S \boldsymbol{q}_j = \sum_{1 \leq i,j \leq n} c_i c_j \lambda_j \boldsymbol{q}_i^T \boldsymbol{q}_j = \sum_{i=1}^n c_i^2 \lambda_i$$

[4] The name comes from physics, in the case when $\boldsymbol{v}$ is the vector of momenta of a system of particles

which is positive (respectively non-negative) if $\lambda_1, \ldots, \lambda_n > 0$ (respectively $\lambda_1, \ldots, \lambda_n \geq 0$). Note: condition (237) is actually equivalent to the matrix S being positive definite (respectively semidefinitie). This gives a simple argument for why S and T being positive definite/semidefinite matrices implies that $\alpha S + \beta T$ is positive definite/semidefinite for any positive numbers α, β.

Remark. *We already know that a big source of symmetric matrices are $S = A^T A$ for an arbitrary matrix A. This matrix is positive semidefinite, because:*

$$\boldsymbol{v}^T S \boldsymbol{v} = \boldsymbol{v}^T A^T A \boldsymbol{v} = (A\boldsymbol{v})^T (A\boldsymbol{v}) = ||A\boldsymbol{v}||^2 \geq 0 \tag{238}$$

Moreover, $S = A^T A$ is positive definite precisely if $A\boldsymbol{v} \neq 0$ for all $\boldsymbol{v} \neq 0$, which happens if and only if A has linearly independent columns (i.e. has zero nullspace). Conversely, any positive definite matrix is of the form $A^T A$ for some matrix A with linearly independent columns.

Let's work out in detail the case of a 2×2 symmetric matrix S, given by (230). Since the eigenvalues of S are given by (231), in order for them to both be positive, we would need $a + c > 0$ and:

$$a + c > \sqrt{(a-c)^2 + 4b^2} \quad \Rightarrow \quad (a+c)^2 > (a-c)^2 + 4b^2 \quad \Rightarrow \quad ac > b^2$$

This is actually equivalent to the stronger property $a, c > 0$ and $ac > b^2$. So we conclude that:

$$\textbf{A } 2 \times 2 \textbf{ symmetric matrix is positive definite precisely when } \operatorname{Tr} S, \det S > 0 \tag{239}$$

As for the energy in the case when S is given by (230), we have:

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = ax^2 + 2bxy + cy^2 \tag{240}$$

It is a well-known fact in algebra that the expression above is positive for all x and y if $a, c > 0$ and $ac > b^2$, precisely as predicted by (239). In fact, you may recognize the equation "(240) = 1" as the equation of a conic in the plane: if S is positive definite, then this conic is an ellipse, while otherwise it's a hyperbola (the intermediary case, when one of the eigenvalues is 0, is a parabola).

Let's do all of this in a numerical example:

$$S = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$$

Its characteristic polynomial is $\lambda^2 - 10\lambda + 9$, whose roots are 9 and 1. The corresponding eigenvectors are $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, hence we conclude that:

$$S = QDQ^T$$

where:

$$D = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} \qquad \text{and} \qquad Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

The ellipses corresponding to S and D are:

$$5x^2 + 8xy + 5y^2 = 1 \qquad \text{and} \qquad 9x^2 + y^2 = 1$$

The latter ellipse has its axes lined up with the coordinate axes. The former has its axes lined up with the eigenvectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$, which is consistent with the fact that going from D to S amounts to rotating the coordinate axes from $\boldsymbol{e}_1, \boldsymbol{e}_2$ to $\boldsymbol{v}_1, \boldsymbol{v}_2$, i.e. applying the linear transformation Q.